AIMS-Rwanda

Algebra and Cryptography- Second assignment

Exercice 1. The following 4 questions are independent:

1. Let N be the product of two distinct primes. Show that N is not a Carmichael number.
2. Let G be a cyclic group of order n. Consider two generators g_1 and g_2 of the group G. Show that $\gcd(\mathrm{dlog}_{g_1}(g_2), n) = 1$.
3. Let p be prime and $n \geq 1$. How many elements of even order do we have in $(\mathbb{Z}/p^n\mathbb{Z})^\times$?
4. Use the Eratosthene's sieve to find the $(5000000 + n)$-th prime number where n is your birthday (mmddyy). How many prime number do we have between 2^{25} and 2^{26}?

Exercice 2. Bob intercepts from Alice the following encrypted message:

$$
\begin{array}{rl}
[[8302588256104991071 3, & 6674026698420872966 1]; \\
\end{array}
$$

$$
\begin{array}{rl}
[[83025882561049910713, & 66740266984208729661]; \\
[117087132399404660932, & 44242256035307267278]; \\
[67508282043396028407, & 77559274822593376192]; \\
[60938739831689454113, & 14528504156719159785]; \\
[5059840044561914427, & 59498668430421643612]; \\
[92232942954165956522, & 105988641027327945219]; \\
[97102226574752360229, & 46166643538418294423]]
\end{array}
$$

To communicate privately, Alice and Bob are using Elgamal cryptosystem. They have choosen the cyclique group $\mathbb{F}_p$ where

$$p = 123456789987654353003$$

and generated by

$$g = 123456789.$$

Their public keys are the following respectively:

$$g_A = 52808579942366933355, \quad g_B = 39318628345168608817.$$

Knowing that Alice and Bob agreed that: Each message consists of a single letter which is encoded as:

$$A = 11, B = 12, \ldots, Z = 36, \text{ space} = 41, \ ' = 42, \ . = 43, \ , = 44, \ ? = 45.$$

can you help Bob to decrypt the message?

Exercice 3.

Create your own public key and private key for Elgamal cryptosystem. Your prime number p has to verify the following:

- $p = 2p_1p_2 + 1$ where p_1 and p_2 are primes;
- $p_1 < p_2 < p_1^3$;
- Its number of digits is not less that 700.

Then, Set up your own ElGamal cryptosystem. Demonstrate how a message addressed to you can be encrypted and how you can decrypt it using your private key.